

This section provides a general overview of Module 4. It includes basic guidelines for preparing the assessment room and conducting the Module 4 administration. The general guidelines are followed by specific instructions for each activity in the module.

Module 4 of the ADOS-2 consists of 10 to 15 activities with 32 accompanying items. Activities are the tasks and interview questions administered by the examiner that allow for a standardized observation of the participant. In contrast, items are the targeted aspects of behavior that the examiner is focused on observing during the administration. Each item is rated by the examiner according to the general coding conventions and the specific coding criteria listed under each item in the Module 4 Protocol Booklet.

Before administering or coding Module 4, it is important to become familiar with all of the items and their accompanying codes. During or immediately following specific activities, the examiner should make written notes about behaviors (e.g., smiling and the voluntary offering of information) that influence coding. Coding should always be completed immediately following the session, based on the live administration.

Module 4 is designed for use with older adolescents and adults who have "fluent" language skills (estimated at a 4-year-old level in functional expressive language or higher) and who have some level of independence in terms of making everyday choices (e.g., what they wear and how they spend small sums of money). With adolescents, examiners may choose between Modules 3 and 4, depending on the participant's interests and level of independence. Module 3 is generally appropriate for adolescents through 15 years of age, whereas Module 4 is generally appropriate for adolescents aged 16 years and older. Some clinicians and researchers may wish to err on the side of administering Module 3 to adolescents when deciding between Modules 3 and 4, because the Module 4 algorithms have not been revised as recently as those of Module 3, and because the Comparison Score is not yet available for the Module 4 algorithm. In addition, some older adolescents may be more comfortable with the structure and visual supports provided by the Module 3 materials.

The activities in Module 4 focus on social, communicative, and language behaviors important in the diagnosis of autism spectrum disorders (ASD) in verbally fluent older adolescents and adults. These activities combine unstructured conversation with a series of structured situations and interview questions that offer a variety of presses for particular kinds of social and communicative behaviors. Some of the activities are designed to provide a relaxed, informal, and unstructured social setting in which the participant has to take the initiative to make social overtures. The goal is to determine how well he or she is able to initiate and sustain a social interaction in situations with a minimum of structure or direction. Some activities in Module 4 are deliberately structured so that the examiner provides a standardized social-communicative task designed to determine how the participant responds to and builds on various types of social stimuli and demands. Other activities comprise specific interview questions about social situations that may provoke particularly strong reactions from some participants.

Fluent Speech

ADOLESCENT/ADULT

1. Construction Task*
2. Telling a Story From a Book
3. Description of a Picture*
4. Conversation and Reporting
5. Current Work or School*
6. Social Difficulties and Annoyance
7. Emotions
8. Demonstration Task
9. Cartoons*
10. Break
11. Daily Living*
12. Friends, Relationships, and Marriage
13. Loneliness
14. Plans and Hopes
15. Creating a Story

**optional*

Module 4 as a whole has three specific goals: (a) to observe the participant's spontaneous social-communicative behavior, given a situation that provides a press to communicate or interact; (b) to assess the participant's ability to carry out particular tasks effectively and appropriately, given the demands of particular situations (e.g., telling a story, teaching a routine skill); and (c) to provide a standard context for the collection of an extensive language sample. Module 4 also offers opportunities to observe the individual's sense of humor and creativity.

The general format of Module 4 creates an interaction that seems natural, and during which planned occasions for certain behaviors arise. Although the goal of Module 4 is to observe natural interactions, many individuals referred for possible ASD—even those who are verbally fluent—have difficulty with long, abstract conversations. Thus, the Module 4 assessment offers the opportunity to alternate standard interview questions with more structured social tasks. The purpose of the interview questions is to observe how the participant converses and the degree of social insight that he or she is able to convey. The structured social tasks are available for two purposes: (a) to provide the participant with a break from highly verbal, often abstract, discussions about sometimes emotionally difficult topics (e.g., "Why did you leave your last job?" or "Do others get annoyed with you?") and (b) to provide presses for particular behaviors that may not necessarily emerge in conversation, such as creative use of materials, use of gesture, and storytelling. The specific social or communicative contexts of each activity in Module 4 are defined by variations in demands, in type of materials, and in the behavior of the examiner.

The heterogeneity within ASD requires that the examiner adapt the ADOS-2 tasks and materials to the level appropriate for each participant, while at the same time providing a social press in each task that is generally comparable across individuals. Thus, if the examiner elects to use Module 4 with an older adolescent but discovers that the participant is particularly interested in materials for one of the tasks in Module 3 (e.g., the action figures), then he or she may wish to switch to Module 3 instead.

The examiner may choose from several alternative sequences when presenting the Module 4 tasks. In general, the order of tasks and events should be flexible and determined by the participant's interests and the flow of the interaction. Many participants with ASD will be more comfortable beginning the session with the more structured tasks (e.g., the "Construction Task," "Description of a Picture," or "Demonstration Task") than entering into a more abstract conversation immediately. When the standard order of presentation is followed, the goal is to move gradually away from the more structured tasks into the open-ended interview questions, which offer an additional, important perspective on the participant's behavior and thoughts. In Module 4's standard sequence, the use of "Creating a Story" as the final task after the interview questions can be particularly helpful for those participants who are most comfortable using materials in a more structured way, as the session then ends on a positive note. Alternatively, this task may be carried out earlier if the examiner feels that it is easier to administer all of the structured tasks consecutively or that it would be more positive to end with the discussion of dreams and future plans.

The interview questions contained in Module 4 fall into two categories. One category of questions is parallel to the socioemotional questions included in Module 3 ("Social Difficulties and Annoyance," "Emotions," "Friends, Relationships, and Marriage," and "Loneliness"). The other category includes additional interview questions unique to Module 4 that ask for specific information about the participant's life and experiences ("Current Work or School," "Daily Living," and "Plans and Hopes"). Some of the additional interview questions, such as those in the "Daily Living" set, primarily provide factual information to facilitate more specific inquiries about the participant's socioemotional functioning. Generally, this information, and the way in which the participant conveys it, impart a sense of the participant's insight into his or her own situation and the social implications of his or her behavior. In addition, all answers offer further opportunities to observe how the participant communicates ideas and feelings. It is also important to recognize that the interview questions in Module 4 are not intended to serve as a psychiatric interview and should not be used in place of standardized interviews for diagnosing other types of disorders.

Sometimes, the participant may be uncomfortable carrying out the more structured, game-like tasks. In most cases, as with any assessment of adolescents and adults, if the examiner offers the activities in a relaxed and positive manner, then the participant is likely to join in without difficulty. Usually, if the examiner demonstrates and shows a willingness to participate in the same task by modeling (except in the case of "Cartoons," where modeling is not allowed) and making encouraging statements, the participant will gradually relax and choose to participate. It is important for the examiner to avoid apologizing for tasks and making the participant self-conscious because of the examiner's own discomfort. If the participant continues to be uncomfortable with any of the more structured tasks despite the examiner's effort, the examiner may want to shift directly to the factual questions regarding the participant's experiences at work or in school, and then proceed through the interview and socioemotional questions before returning to the structured tasks. The structured tasks offer a unique perspective on the participant's ability

to carry out different aspects of social interaction, so it is important to administer all of the required activities. If the participant categorically refuses to carry out a task and cannot be convinced to try it, it is best to move on to the next task. However, the examiner must not allow any personal discomfort or inexperience to result in his or her omitting certain tasks or unintentionally increasing the participant's self-consciousness.

The ADOS-2 differs substantially from other types of assessments, such as cognitive testing or psychiatric interviews. The session as a whole is primarily designed to assess social and communication behaviors; therefore, it is crucial to prevent the session from resembling a clinical interview. The manner in which a participant responds to a question or task is as important as the content of his or her response. It is essential from the outset to provide an interactive model, commenting on the participant's activities or statements and introducing brief observations about the examiner's own interests or activities. Many participants will expect a more directive question-and-answer approach from an adult authority figure, and the examiner must counteract this by adopting an informal, personal, interactive style that makes use of creativity, imagination, and humor. Generally, a psychotherapeutic style, in which the examiner interprets the participant's comments back to him or her, is not helpful in this context unless the participant is becoming distressed, in which case topics should be changed and the participant comforted. The examiner also should avoid a task-oriented style (as in a cognitive assessment).

For each activity, there is a specified focus of observation with respect to the participant's behavior in that task. Ratings are completed at the end of the assessment on the basis of the participant's behavior during the Module 4 administration. The examiner should be thoroughly familiar with these ratings so that they may be kept in mind throughout the session. It is not appropriate to read the ADOS-2 Manual during the administration of the assessment. If the examiner has not had sufficient practice to work from the Protocol Booklet and brief notes, the administration cannot be considered valid.

Some activities in Module 4 are optional. The examiner decides to administer the optional activities depending on whether (a) the examiner feels that more background is needed (e.g., the "Daily Living" questions), (b) the participant needs the opportunity to answer in a positive way after a series of difficult questions (e.g., those included in "Current Work or School"), or (c) a sufficient sample of language and gestures has not yet been obtained (e.g., "Description of a Picture" or "Cartoons" activities). It is generally advisable to administer all of the Module 4 activities whenever possible, as coding can be difficult or invalid if sufficient information has not been obtained from the activities administered.

When deciding whether to administer the optional interview questions in particular, it is important to remember that some of the optional questions provide the primary opportunities to elicit certain information necessary for coding (e.g., the "Current Work or School" and "Daily Living" tasks usually influence the "Responsibility" item). The optional questions also address topics (e.g., money, leisure activities) that persons with ASD are likely to find interesting and easier to discuss than some of the more social questions. The examiner is not required to administer the optional interview questions if the information has already emerged through earlier conversation, or if the examiner feels certain that the overall ratings can be made on the basis of the required activities alone. Nevertheless, the optional interview questions are often needed in order to obtain all of the information necessary for coding, and the optional questions provide a break from the more difficult questions that may be uncomfortable for the participant. It is preferable to administer both the required and optional activities in Module 4 whenever possible.

Figure 6 lists the standard set of materials used with the activities in Module 4. Note that there are no materials for any of the interview questions. The materials of the ADOS-2 have been selected specifically for individual tasks and presses. The materials for Module 4, in particular, have been deliberately selected to be interesting, humorous whenever possible, and appropriate for adolescents and adults. Examiners should avoid introducing additional materials that are not included in the standard administration. Additional materials may be used to motivate the participant if absolutely necessary, but they should never be used as alternative stimuli for standard tasks. Conducting the assessment in a person's home may make it more difficult to control the assessment environment or to prevent the introduction of extraneous materials. When it is necessary to complete the ADOS-2 administration in the home setting, care should be taken in figuring out where to work and how to lay out materials.

Preparing the Assessment Room for the Module 4 Administration

- Pay close attention to the physical structure of the setting, including the available furniture and space, to select an arrangement that is most suitable. In order to provide encouragement for informal back-and-forth interactions and communication, it is generally most appropriate to have an arrangement in which you and the participant are facing each other diagonally at the corner of the table or at a round table. It is preferable to

Activities	Materials needed
1. Construction Task (<i>optional</i>)	puzzle pieces and printed design to be duplicated
2. Telling a Story From a Book	2 picture storybooks for Modules 3 and 4
3. Description of a Picture (<i>optional</i>)	U.S. map scene and resort scene
4. Conversation and Reporting	—
5. Current Work or School (<i>optional</i>)	—
6. Social Difficulties and Annoyance	—
7. Emotions	—
8. Demonstration Task	hand towel and soap (if necessary)
9. Cartoons (<i>optional</i>)	Series A: fisherman/pelican series Series B: monkey/coconut series
10. Break	mini game (e.g., shape puzzle), drawing paper, set of 8 markers, pin art, spin pen (spin top with pen base), small radio, current newspaper and magazine
11. Daily Living (<i>optional</i>)	—
12. Friends, Relationships, and Marriage	—
13. Loneliness	—
14. Plans and Hopes	—
15. Creating a Story	6 small objects with a definite purpose (e.g., umbrella, car), 6 small objects with no clear purpose (e.g., piece of string, wooden block)

Note. Some of the materials listed in this figure are also used in other ADOS-2 modules.

Figure 6
Materials Used With Module 4

avoid an arrangement in which you and the participant are seated opposite one another across a table, as this arrangement may feel confrontational to the participant.

Beginning the Module 4 Administration

- Begin the assessment with an attempt to put the participant at ease. This should involve comments of greeting and some indication that the session will involve a variety of activities and conversations. Include general chat and an expression of interest in the participant and his or her activities if this will help put him or her at ease.
- Keep the introductory period short in order to move on to a more structured task, such as the "Construction Task" or "Description of a Picture." The provision of some structure and focus in these tasks serves to put most individuals with ASD or other developmental disabilities at greater ease than they might feel if expected to participate in unstructured conversation.

During the Module 4 Administration

During the Module 4 administration, you must structure the interaction. The following guidelines are helpful:

- Aim for a standardized administration of 40 to 60 minutes.
 - Module 4 often involves more conversation than the other ADOS-2 modules, so the administration time for Module 4 is generally longer than the time for the other modules. The administration should not typically last longer than 60 minutes, however, even when administering all of the optional activities.
 - If the participant provides leads that may be important clinically (e.g., reports significant depressive symptoms), respond with a genuine but brief show of empathy and support, assuring the participant that those leads will be followed up as soon as the activities are completed. Remember to return to those topics as soon as you have finished the ADOS-2 administration, but do not include any of this follow-up information in your coding considerations.
- Take care to encourage and, to some extent, structure the task-oriented aspects of the assessment, without directly prompting specific behaviors. Your ability to carry out the administration in this style is critical in allowing the participant to exhibit his or her own level of social behavior.

- Refrain from providing so much help that there is no opportunity to observe the participant's spontaneous behavior. In contrast, avoid providing so little structure that the participant is uncomfortable.
- Be both active and interactive in order to produce the appropriate presses for social-communicative interchanges. Nevertheless, except when specified otherwise, codes are primarily based on the participant's initiatives in making, using, and developing these opportunities, rather than on his or her reactions to your behavior, creativity, or use of materials.
- Do not constantly strive to sustain the interaction. The participant must be provided with some clear opportunities to initiate and maintain interaction. For example, you may sometimes need to be quiet, or allow an "awkward silence" to persist for a moment or two.
- Take care to avoid coding stereotyped or imitative responses as showing creativity. The ADOS-2 consists of relatively brief observations, however, so expectations for extraordinary originality should not be high.
- Use opportunities throughout the session to assess the participant's language skills and style of communication. Follow the participant's lead whenever appropriate. The participant's communication style is much better evaluated on the basis of language production during a back-and-forth conversational interchange than through question-and-answer cross-examination. This means that the conversation phase of the Module 4 assessment can be introduced earlier or later than indicated in the standard sequence of activities. Conversational interchanges can (and should) also occur throughout the other activities, not only during the specified "Conversation and Reporting" task. The opportunities provided in the structured tasks are also very important to this observation.
- Give appropriate encouragement and praise throughout the session, and clearly share enjoyment in the participant's activities.
- Observe whether the participant shows any repeated unusual or inappropriate behaviors or social disinhibition (e.g., touching your hand, picking his or her nose). If so, make an overt attempt at some point to ask the participant to stop, or you should try to prevent these behaviors. For example, interrupt the participant while he or she is providing a long list of dates. This attempt will enable you to determine how much the participant's behaviors interfere with the interaction, as well as how the participant responds when he or she is interrupted.
- Note that some of the tasks are arranged so that there are two or more opportunities for the participant to show the behavior to be coded. For example, the "Description of a Picture" activity includes two potential pictures. In general, it is not necessary to provide the additional opportunities if the participant has produced the behaviors that are the focus of observation during the first opportunity.
- Administer both the required and optional activities in Module 4 whenever possible. The optional activities (both the materials-based tasks and interview questions) are often needed to obtain all of the information necessary for coding, and they provide a break from the more difficult questions that may be uncomfortable for the participant.

Instructions for Activities in Module 4

This section includes specific instructions for each activity in Module 4. The specific activity instructions include the purpose of the activity, the materials needed, administration instructions, and the focus of the observation. Within each set of activity instructions, verbal directions to the participant are written in **bold**.

1. Construction Task (*optional*)

...the activity and an opportunity to observe the participant's interactive

Materials: Puzzle pieces and printed design to be duplicated.

Instructions:

- Remove all of the puzzle pieces from their container, and place them on the table so that the participant can see them but cannot reach them. It is essential that you sit sufficiently far apart from the participant so that some of the blocks can be placed outside of the participant's reach.
- Place the printed design in front of the participant, point to the puzzle pieces and the design, and say, **"I want you to put these pieces together to fill in this picture. You are supposed to use all the pieces. You can start with these [hand a few pieces to the participant]. Let me know when you need more pieces."**
- Keep the remaining pieces on the table within sight but outside of the participant's reach and on the other side of your arm. Gesture to indicate the pieces you have kept and ensure that the participant can see these additional pieces. When the participant has assembled the few pieces he or she was given, wait to see whether the participant will ask for access to the remaining pieces. If so, provide the participant with a few additional pieces at a time. If the participant does nothing, deliberately look at him or her.
- Observe the participant's response to your direct gaze. If the participant makes no response to your direct gaze in 3 to 5 seconds, gesture toward the participant's design and say, **"Are you doing alright?"** or **"How are you doing?"** If the participant makes no response to your general prompt, then you can use a more specific prompt by asking, **"Do you need more pieces?"**
- Continue the task until all the puzzle pieces are in place. When the task is finished, place the container for the puzzle pieces in front of the participant, and open it. Then, put some of the pieces in the container while saying, **"Let's move on to something else."** Watch to see whether the participant helps to put away the remaining pieces.

Special administration notes: You should ensure that the task does not take more than a few minutes and is not frustrating for the participant because of visual-spatial or motor demands. For this reason, two versions of the design, one with and one without lines demarking each piece, are provided on alternate sides of the printed design form. In addition, if the participant seems unable to complete the design, you may complete most of the "puzzle" for him or her and ask him or her to finish it.

Focus of observation: Observe and evaluate whether the participant indicates the need for more pieces, and if so, how he or she attempts to do so. (For example, does he or she reach over your arm? Does he or she vocalize, gesture, or make eye contact?)

2. Telling a Story From a Book

Purpose: This activity assesses the participant's ability to recount a sequential story from a book of pictures and to provide a context for comments about social relationships and affect.

Materials: Either of the two picture storybooks from the materials for Modules 3 and 4 may be used. Two books are provided in case one is not appropriate for the participant (e.g., the participant has recently seen one of the books).

Instructions:

- Present one of the books to the participant and say, **"Let's look at this book. It tells a story in pictures. See, it starts out with... [describe the first picture in the book]. Can you tell me the story as we go along? You go first. Then, I'll take a turn."** This instruction is provided to help the participant orient to the book's story and to warn the participant that you may interrupt him or her.
- Hand the book to the participant, giving whatever prompts may be necessary to encourage him or her to start describing the story and then to turn the page. If the participant seems hesitant or reluctant, you should give *no more than two* specific prompts to get him or her started. For example, you might describe the action in one picture and say, **"I wonder what happens next,"** and then give the participant a chance to comment. If the participant focuses on specific details of individual pictures, you should say, **"You're right. Can you tell the story?"** or **"What are the pictures all about?"** (Make *no more than two* such prompts.)
- *Do not model labeling emotions.* For example, do not make comments such as "Oh look, this frog is really surprised." One of the focuses of this task is to observe whether the participant labels or comments on the emotions of the characters in the book. Therefore, the participant must have the opportunity to do so spontaneously.
- Allow the participant a few minutes to describe the book. Then say, **"That was great. Now I'll take a turn,"** and quickly complete the story for or with the participant. More back-and-forth turns can occur if you need to speed

or slow the participant's pace through the story to prevent either the task taking too long or the participant rushing through the task without sufficient opportunities to observe behaviors. This also allows you to observe whether the participant will allow someone else to take a turn telling the story. If the participant is determined to complete the story, note this, and let him or her do so.

Special administration notes: If the participant comments that the book seems "childish," you can call attention to some of the more creative aspects of the story (e.g., figures of speech) and eventually initiate a conversation around books the participant likes to read.

Focus of observation: Obtain a sample of the participant's spontaneous language and communication, as well as a sense of what captures his or her interest. In addition, evaluate the participant's response to conventional humor, his or her spontaneous comments about how the characters in the story are feeling, and the degree to which he or she can convey continuity in a story.

3. Description of a Picture (optional)

Purpose: This activity provides an opportunity to generate a sample of language and/or other communicative behaviors.

Materials: U.S. map scene and resort scene. Both pictures should be available, but generally only one is used for each participant. The simpler feast scene from Module 2 may be used as an alternative, if it is more appropriate to the participant's developmental level or interests.

Instructions:

- Tell the participant, **"Let's look at this picture now. Can you tell me about it? What is happening here?"** The aim is to generate language, so encourage the participant to comment about the picture, and respond positively to what the participant says.
- Be enthusiastic and interested in what the participant says, but do not provide further substantive information about the picture that is being described. For example, if the participant points to the sailboat and says, "That looks like a sailboat," you might say, **"I've never been on one. Have you?"** If the participant indicates the man on the surfboard and says, "Look, he's near the Hollywood sign," you might say, **"You're right" or "Yes, he is. Have you ever been to California?"**
- Note whether the participant relates the picture to his or her own experiences (family, friends, activities, and so on). If so, encourage this opportunity to generate conversation. The conversation can be evaluated when codes are assigned at the end of the assessment.
- Comment and use a variety of general questions and responses throughout the activity. Refrain from modeling or asking specific questions about particular parts of the picture.
- Allow the participant time to comment on the picture spontaneously and to respond to general prompts and questions as just described. If the participant does not respond to general prompts and questions, provides only isolated specific labels of objects or people, or mentions only one statement containing an agent, object, or action, then you should model the use of more complex language. For example, you might say, **"Look over here. That man on the tractor must be a farmer. He's next to a big ear of corn."** If the participant has not made any such statements or identified any objects in the picture, then you should ask specific questions: **"What's this?" "Who is this?" "What is he doing?" or "What is happening over here?"** Continue to provide encouragement and amplification of the participant's answers.
- Present the second picture only if a sufficient language sample is not obtained with the first picture or if the participant shows no interest in the first picture.

Focus of observation: Obtain a sample of the participant's spontaneous language and communication, as well as a sense of what captures his or her interest.

4. Conversation and Reporting

Purpose: This activity assesses the participant's ability to engage in a conversation with back-and-forth interchanges and to describe an event or situation that occurred outside of the immediate context. In addition, this activity evaluates the participant's ability to recount a nonroutine event (e.g., a birthday party, a vacation, a particular interaction with a person at school or work). This task also provides an opportunity to generate a language sample in less structured circumstances than are provided by the picture tasks.

Materials: None.

Instructions:

- Use necessary judgment throughout the evaluation to determine what is most likely to set the participant at ease and to invite conversational interchange. Ordinarily, the conversational phase works best if you start with talk about what has happened during the session or about what the participant has said. At an appropriate time, develop the conversation to discuss topics outside the immediate situation (e.g., school or work, siblings, leisure activities) in order to assess the participant's ability to report on situations or events that have happened in the past.
- Use earlier opportunities for conversation during the session if this seems appropriate. For example, conversation about the participant's interests in reading novels or watching movies could be a topic that follows the activity "Telling a Story From a Book."
- Avoid a question-and-answer-style interaction, instead making a deliberate effort to use a conversational approach in responding to the participant, commenting on what he or she says and developing the discussion further.
- Show interest and involvement in the conversation and respond to the participant's talk with appropriate "hand-over" prompts in order to provide a press for a back-and-forth interchange. To emphasize the reciprocal nature of the task, you should provide conversational leads—for instance, by making brief statements about your own interests, activities, or feelings—and then wait to see whether the participant can build upon or follow up on such comments.
- Provide an opportunity for the participant to describe a nonroutine event (e.g., a vacation or family celebration). This event should be something that actually occurred, as opposed to an account of a film or story. For example, a report about a vacation or family celebration may follow from "Description of a Picture." Specific questions may be needed to introduce a topic. At some point, however, you should stop maintaining the conversation and remain silent for a few seconds while looking interested, to see whether the participant can take the initiative to recount an event without a specific prompt.

Special administration notes: In order to observe whether the participant can carry on a conversational interchange, you must provide sufficient leads, guides, and prompts on topics. Many individuals with ASD become very inhibited if presented with an unstructured situation that is too open, and in which they have to take the lead with an unfamiliar authority figure. It is, therefore, important for you to make appropriate use of the participant's own interests in such an interchange, by incorporating the participant's earlier statements, comments, or questions whenever possible. Conversations can begin about a participant's strongest interests but should not be centered exclusively on preoccupations or specific favorite topics (e.g., power tools). The conversation should include some discussion of age-appropriate topics of interest (e.g., music, pets, outings, sports) that are not stereotyped or circumscribed.

Focus of observation: Evaluate the extent to which the participant builds on your statements, elaborates on his or her own statements to provide leads for you, and takes a full role in back-and-forth conversation, particularly about a topic outside of the immediate context. Pay particular attention to how the participant reports *routine* and *nonroutine events* and how he or she describes *relationships* and *emotions*. Also, observe features of the participant's communication, including his or her use of gaze, facial expression, intonation, and gesture.

5. Current Work or School (optional)

Purpose: These questions evaluate how the participant describes his or her current situation, his or her level of insight about everyday situations, and whether he or she understands his or her role in determining what will happen in the future. These questions also sometimes elicit information relevant for coding "Responsibility" and "Insight Into Typical Social Situations and Relationships" (e.g., the relationship between an employer and an employee, or that between a student and a teacher).

Materials: None.

Instructions:

- Ask the "Current Work or School" interview questions in the order listed here.
- Omit the employment questions, and proceed to the questions about school, if the participant is in school full time or does not have a job and has never been employed.

Interview questions about work:

- "Do you have a job?"

If so:

- “What kind of job is it? How did you find it?”
- “Have you had other jobs before?”
- “Are you happy where you are or would you like to move on to something else eventually? What would it be?”
- “What about your co-workers? Do they seem to be happy or are they ready to move on?”

If not:

- “What do you do during the day?”
- “Did you have a job before? Why did you leave your old job? Was it something you had planned?”
- “Would you like to have a job someday?”

If yes:

- “What would it be?”
- “What will you need to do to find this type of job?”

If no:

- “What would you like to do?”

OR

Interview questions about school:

- “Are you in school? Where?”
- “What courses are you taking?”
- “What year (grade) are you in? How is it going?”

If the participant is no longer in school and not employed:

- “How far did you go in school? How did it go?”
- “What do you plan to do next? What experience/training would you need to do that?”

Focus of observation: Obtain general information about whether the participant is employed and/or in school, whether he or she has had to leave jobs, and if so, for what reasons. Evaluate whether the participant has a realistic understanding of the possibilities for future employment. In addition, determine what the participant's school experience is or was like, and explore whether he or she has a realistic understanding of the training or experience necessary for future employment.

6. Social Difficulties and Annoyance

Purpose: These interview questions assess the participant's insight into personal social difficulties and sense of responsibility for his or her own actions.

Materials: None.

Instructions:

- Ask all of the “Social Difficulties and Annoyance” interview questions in the order listed here.

Interview questions:

- “Have you ever had problems getting along with people at school or at work? How about at home? Do you ever get in trouble? Why? What for?”
- “Are there things that other people do that irritate or annoy you? What are they?”
- “What about things you do that annoy others?” (If no response, ask: “What about your family members?”)
- “Have you ever been teased or bullied? Why, do you think?”
- “Have you ever tried to change these things? Have you ever done anything so that others wouldn't tease you? How has it worked?”
- “Are there other people you know who get teased or bullied?”

Focus of observation: Evaluate the participant's perception of social difficulties, his or her insight into the nature of these problems, and whether he or she has made any attempt to change his or her own behavior in order to fit in with others more smoothly. Pay attention to the participant's understanding of the appropriateness and implications of his or her feelings and behaviors.

7. Emotions

Purpose: This task provides a context in which the participant talks about emotions and personal experiences. The goal is to observe both the content of what the participant says and also the degree to which the participant who is verbally fluent can use language to talk about more abstract topics, particularly those associated with emotions. The requirement to talk about how one feels during different emotions is intended as a press to talk about a topic that is a challenge for many people. It is not intended as a test of cognitive function, so if it is very difficult for the participant, you should note this and continue to the next probe.

Materials: None.

Instructions:

- Introduce the task by saying, "Now I'd like to ask you a few questions." *Do not introduce the task as an "emotions" task.*
- Ask the "Emotions" interview questions in any order you choose. It is often appropriate to begin or end with "happy" or other positive emotions.
- Continue to ask the "Emotions" interview questions until the participant has given detailed descriptions of at least *two* emotions, including the contexts in which they arise and the nature of the participant's individual experience of these emotions. You need not ask about all emotions if the participant offers vivid and clear descriptions of emotions.
- Do not remain on one emotion probe for too long. Do not "insist" on a good response before moving on; many participants may not be able to provide detailed descriptions of emotions.
- End the task after the participant has provided enough detail (as just described) or after all of the "Emotions" interview questions have been asked. If you are uncomfortable ending this segment with questions about negative emotions, you may want to follow up with questions about anticipating something pleasant or about feeling relaxed and peaceful.

Interview questions:

- "What do you like doing that makes you feel happy and cheerful?"
- "What kinds of things make you feel this way? How *do* you feel when you're happy? Can you describe it?"
- "What about things that you're afraid of?"
- "What makes you feel frightened or anxious? How does it feel? What do you do?"
- "What about feeling angry?"
- "What kinds of things make you feel that way? How do you feel 'inside' when you're angry?"
- "Most people have times when they feel sad. What kinds of things make you feel that way?"
- "How *do* you feel when you're sad? What is it like when you're sad? Can you describe that?"
- "How about feeling relaxed or content? What kinds of things make you feel that way?"

Focus of observation: Identify the events or objects that elicit different emotions in the participant, particularly whether they are social in nature or not. In addition, observe how the participant describes his or her emotions. Also, observe whether the participant exhibits facial expressions or creative uses of language in the context of describing his or her emotions and others' emotions, and whether he or she displays insight into typical social relationships that may cause some of these emotions.

8. Demonstration Task

Purpose: This activity assesses the participant's ability to communicate about a familiar series of actions using gesture or mime with accompanying language, and to report on a routine event. In this task, the participant is

toothbrush." Then, pretend to draw the toothbrush. Say, "This is the toothpaste." Then, say, "And this is a cup." Again, indicate the pretend objects. These gestures should be slow and without extraneous movements.

- Avoid acting out any aspects of the task (e.g., turning a faucet on) unless necessary to get the participant started; this is important because the participant cannot receive credit for using a gesture that you already demonstrated.
- Reiterate, "Now I want you to *show me and tell me how to brush my teeth. Start right at the beginning. You've just come into the bathroom to brush your teeth. What do you do now?*"
- Observe the participant's demonstration. If the participant demonstrates an isolated action very rapidly and conventionally, or pantomimes without speaking, say, "That's great. Now, *tell me and show me again right from the beginning—from when you first came into the bathroom.*" Similarly, if the participant talks without producing any gestures, say, "That's great. Now can you try it again, but this time *show me and tell me at the same time.*"
- Determine whether the participant has understood the task. If the participant does not understand the idea of the task, describe the scene again in the same manner. If the participant continues to demonstrate a clear lack of understanding of the task (i.e., if the participant stares blankly and does not attempt the task), you may demonstrate (in very simple terms) a different event (e.g., using words and gestures to demonstrate driving a car by first putting the key in, turning on the engine, and holding the steering wheel) before proceeding to the alternative task of handwashing. *Note that the demonstration of a different event should occur only if the participant does not understand the task; you should not demonstrate just because the participant fails to adequately integrate gesture and language.*
- Proceed to handwashing. Conduct this trial in a fashion similar to the toothbrushing event, this time setting the scene explicitly by pointing out the following *pretend* objects: soap, a hand towel, and a sink with faucet handles.
- Say, "Now I want you to *teach me how you wash your hands using soap.*" If the participant still does not seem to understand the task, reset the scene once more, using the real soap and towel with the pretend sink. These real materials are included primarily to end the activity on a successful note for those participants who are unable to act out or explain a routine event.

Focus of observation: Determine whether and how the participant represents familiar actions in gesture, particularly through the use of his or her body to represent an object (e.g., a finger for a toothbrush) or in miming the use of a pretend object. In addition, use this activity as an opportunity to evaluate the participant's report of a routine event and the pragmatics of teaching a sequence of actions.

9. Cartoons (optional)

Purpose: This task provides an opportunity to observe the way in which the participant narrates a story, uses gesture to enact events, and integrates gesture with gaze and language. It also provides an opportunity for commenting on the emotions of the characters in the stories.

Materials: The *Series A* cartoons (a story involving a fisherman, a cat, and a pelican) and the *Series B* cartoons (a story about two monkeys and some coconuts). Each set of cards presents a brief story in cartoon form, one frame per card, without any dialogue or narrative text.

Instructions:

- Tell the participant that he or she will now be shown some cartoons and then will be asked to retell the story depicted in the cartoons without looking at the pictures.
- Choose either Series A or Series B, and offer a brief statement describing the setting of the cartoons in very general terms (e.g., "This is a story about a fisherman and a cat" or "This is a story about two monkeys").
- Look through the cards one by one with the participant, encouraging him or her to comment on what is happening in each card. Try not to give much information about the cards. If the participant is confused about what is happening in the story, however, you may help clarify the events depicted.
- *Do not model labeling emotions.* As with "Telling a Story From a Book," the participant must have the opportunity to label emotions spontaneously.
- Ask the participant to push his or her chair back from the table (so that the table is not within reach) and then to stand up and tell the story. It is often helpful if you stand up, too. It is important that the participant is not holding anything in his or her hands or touching the table, chair, or wall, and that his or her chair is pushed back

from the table so that he or she will be able to gesture freely. If an observer is in the room, it may be helpful to ask the participant to tell the observer the story.

- Note whether the participant seems hesitant or uncomfortable.
 - If the participant seems hesitant to stand and retell the story, it can be helpful to place two pieces of paper on the floor in appropriate places, telling the participant to stand on one piece of paper as “the stage” while you stand on the other piece, signifying “the audience.”
 - If the participant is markedly uncomfortable standing to retell the story, he or she may be permitted to remain seated; however, you should make a concerted effort to get the participant to stand. In the event that he or she insists on staying seated, attempt to get the participant to pull his or her chair away from the table, as most people are much more animated telling stories if they cannot prop their arms and hands on a table.
- Try to prevent the participant from looking at the story cards while retelling the story. Although participants will often ask to do so, it is important to prevent them from doing so whenever possible because individuals are much less likely to gesture when looking at the story cards. If the participant appears to be particularly distressed about it, however, you can say, **“Give it a try first, and I’ll hold up a card to help you remember if you really need it.”** Remember that the “Cartoons” activity is *not* a test of the participant’s memory of the story but rather is intended to provide information about how the participant uses various modes of communication to retell a story.
- Proceed to the second series of cartoons only if the first series is “spoiled” in some way (e.g., the participant puts his or her hands in his or her pockets during the retelling but seems too anxious to be interrupted, or he or she insists on looking at the cards). In this event, ask the participant to retell the second series of cartoons using the procedures just described for the first series.

Focus of observation: Observe the participant’s use of gesture and its coordination with speech, as well as his or her response to humor. Also, use this task to obtain an additional language sample from the participant and a sense of his or her degree of flexibility in adapting a narrative to the audience of the listener. In addition, note any comments the participant makes about emotions and relationships.

10. Break

Purpose: The primary purpose of this activity is to observe the participant’s behavior in less structured circumstances. A secondary purpose is to give the participant a break from the social demands of the assessment.

Materials: Mini game (e.g., shape puzzle), drawing paper, set of eight markers, pin art, spin pen (spin top with a pen base), small radio, and current copies of a newspaper and a magazine. A snack may be provided, if necessary.

Instructions:

- Determine an appropriate time for the “Break” activity to occur. The break may be given at any time during the assessment. Several breaks may be taken, if necessary. The materials for the break may be presented at the very end of the Module 4 assessment if it seems that the participant would be most comfortable completing the more structured activities first.
- Say, **“Let’s take a break,”** and indicate that you need some time to catch up on your notes.
- Point to the specified “Break” materials, and express the hope that the participant can find something of interest among them. If the participant is unfamiliar with any of the materials, you should demonstrate how they work (e.g., take the cap off the spin pen, put your hand in the pin art).
- Say, **“I need to make some notes for a few minutes,”** and move your chair back from the table so that you are sitting within view but away from the table where the participant is sitting.
- Note the participant’s interest in the “Break” materials. If none of the provided materials interests the participant, ask the participant if there is something else he or she would like to do, or offer a snack and/or something to drink. Once the participant is settled and you have made a reasonable attempt to interest him or her in the materials (even if nothing is of interest), you should work on your notes for at least 2 minutes (slightly longer is fine if helpful to you or the participant).
- Observe whether the participant initiates an interaction. If so, respond briefly and positively, but indicate that you have to finish more paperwork before you can talk. After a few seconds, look up, catch the participant’s eye, and smile briefly in encouragement. The goal is to create an occasion for the participant to initiate an interaction. If this does not occur, you can return to your notes or say, **“I’ll just be another minute.”**

- Return to the table after a few minutes total. Make it clear that you are finished taking notes (e.g., pull your chair back up to the table, put your pen and Protocol Booklet down). Wait to see whether the participant invites you back into the interaction by making a social overture. If necessary, you can look at one of the objects on the table, indicating interest in it or what the participant is doing with it, but continuing to allow the participant to take the lead.

Special administration notes: This task should be completed with the "Break" materials as outlined. If the participant mentions a preference to leave the room during the break, you should clarify that this will be a very short break within the room. You can offer that a longer break will be possible soon (if needed). If a true break outside of the room is necessary (e.g., to use the restroom or take medication), be sure to administer the "Break" task as outlined here when the participant returns.

Focus of observation: Observe how the participant occupies himself or herself during free time, and how he or she responds to your withdrawal from and return to the interaction. Also, observe whether and how the participant initiates and participates in an unstructured conversation or interaction with you at the end of the break.

11. Daily Living (optional)

Purpose: These questions can be used to obtain factual information and background for the socioemotional questions and to evaluate the participant's understanding and views regarding money, residential arrangements, and leisure activities. These questions are particularly useful in evaluating the degree to which the participant is responsible for himself or herself in activities of daily living, as well as whether the participant sees himself or herself as being responsible for his or her own actions.

Materials: None.

Instructions:

- Ask the "Daily Living" interview questions in the order listed here.

Interview questions about money:

- "What about money? Do you take care of your own money?"
- "Where does it come from? Who pays your bills?"
- "Have you ever saved your money to buy something or do something special? What was it?"

Interview questions about residential arrangements:

- "Where are you living now?"

If living at home with parents:

- "Have you ever lived away from your parents?"
- "What would be different about living on your own?"
- "Would you like it better?"
- "What would be difficult?"

If living on his or her own:

- "How did you find the place where you live now?"
- "Who do you live with?"
- "Can you tell me a little about it?"

Interview questions about leisure activities:

- "What do you like to do in your spare time at home?"
- "What about going out?"

Focus of observation: Obtain factual information about the participant's level of financial responsibility. Also, establish where the participant is living, and how this arrangement was made. In addition, determine the extent to which the participant is realistic about plans for independence and the complexities of various living arrangements.

12. Friends, Relationships, and Marriage

Purpose: These interview questions provide an opportunity to obtain a detailed description of one or more relationships that the participant would consider to be friendships and also to obtain a general description of his or

her understanding of the concept of friendship and the idea of getting married or building a long-term relationship. Note that the purpose of this activity is *not* to determine whether the participant has friends but rather to evaluate how he or she understands the concepts of friendship and marriage/long-term relationships.

Materials: None.

Instructions:

- Ask all of the "Friends, Relationships, and Marriage" interview questions regardless of a participant's chronological age. The questions do not necessarily need to be read verbatim.
- Make appropriate modifications to the language of the questions depending on the developmental level of the participant. For example, if the participant does not have any experiences with or is reluctant to talk about a girlfriend/boyfriend, you could ask **"Do you know of anyone who has a girlfriend or boyfriend?"** and ask **"How do you know she/he is his/her girlfriend/boyfriend?"** Note that participants of either sex should be asked the question as worded: **"Do you have a girlfriend or boyfriend?"**
- Be sure to maintain the "spirit" of the questions and proceed through all of the questions in order to ensure that the information necessary for coding is elicited.

Interview questions:

- **"Do you have some friends? Can you tell me about them?"** (Note the ages of the friends. It can be helpful to ask for names if the participant is very general about who his or her friends are. Sometimes the names provided are family members or professionals or a list of an entire class, which is useful to know in terms of interpreting the answers to the questions about friendship.)
- **"What do you like doing together? How did you get to know them? How often do you get together?"**
- **"What does being a friend mean to you? How do you know someone is your friend?"**
- **"How is a friend different from someone whom you just work with or go to school with?"**
- **"Do you have a girlfriend or boyfriend? What is her/his name? How old is she/he?"**
- **"When did you see her/him last?"**
- **"What is she/he like? What do you like to do together?"**
- **"How do you know she/he is your girlfriend/boyfriend?"**
- **"Where do you want to live when you get older? What kind of place (apartment, house, condo)?"**
- **"Whom do you think you would like to live with? Your family, a roommate(s), by yourself?"**
- **"Do you ever think about having a long-term relationship or getting married (when you are older)?"**
- **"Why do you think some people get married or live with a girlfriend or boyfriend when they grow up?"**
- **"What would be nice about it? What might be difficult about being married or living with a girlfriend or boyfriend? Or living with a roommate?"**

Focus of observation: Evaluate how the participant understands the concepts of friendship, marriage, and other social relationships, and the nature of these relationships. Also, evaluate the participant's understanding of why a person might want to be involved in a long-term relationship and of his or her own possible role in such a relationship.

13. Loneliness

Purpose: These interview questions provide another opportunity to assess the participant's insight into his or her social situation, as well as the participant's ability to describe his or her emotional reaction to it.

Materials: None.

Instructions:

- Ask all of the "Loneliness" interview questions in the order listed here.

Interview questions:

- **"Do you ever feel lonely?"**
- **"Do you think other people your age ever feel lonely?"**
- **"Are there things that you do to help yourself feel better? What about things other people do to help themselves feel better when they're lonely?"**

Focus of observation: Evaluate whether the participant understands the concept of loneliness and how he or she feels it pertains to him or her or to other people.

14. Plans and Hopes

Purpose: This question gives the participant an opportunity to describe any goals or aspirations that he or she may have; it is also intended to provide a positive end to the interview. This question is also often useful in obtaining information relevant for coding "Responsibility."

Materials: None.

Instructions:

- Ask the "Plans and Hopes" interview question as listed here.

Interview questions:

- "Do you have plans or dreams for the future of things that you would like to do, or have, or see happen to you? Tell me about them."

Focus of observation: Obtain information about what the participant anticipates in the future. Try to end the interview on a positive note.

15. Creating a Story

Purpose: This task provides an opportunity to observe creativity in a play-like situation that is appropriate for verbally fluent children, adolescents, and adults. The participant's task is to use five objects to make up a story, newscast, or commercial.

Materials: Six small objects with a definite purpose (e.g., umbrella, car) and six small objects with no clear purpose (e.g., piece of string, wooden block).

Instructions:

- Tell the participant, "Now you and I are going to make up stories using some of these objects. First I will pick five things and use them to make up a story, and then you will have a turn to make up a story using five different things."
- Model choosing five objects and making up a simple story. Use three objects in a creative or novel way (e.g., using a piece of sponge as a bed, using an umbrella as a basket) and two objects in the way that they were intended to be used (e.g., using a paper clip as a paper clip). Use one object as the "actor" in the story (e.g., "Mr. Flame woke up one morning," using a candlestick). After you have modeled the creation of a story, place your five objects off to the side.
- Gear your story to the developmental level of the participant. Make it simple enough so as not to make the participant feel intimidated or overwhelmed by the task.
- Encourage the participant to select five new objects from the bag to use in making up a story of his or her own. It is generally advisable to leave the car for the participant to use, since this is a relatively easy object to incorporate into a story. Make sure that the participant does not select only "real" objects, however, so that he or she is forced to use at least some of the more abstract objects (i.e., "junk" objects) in creating his or her story. An easy way to do this is to be familiar with the objects in the "Creating a Story" bag such that using two or three "real" objects in your own story will leave only two or three "real" objects for the participant to choose.

Focus of observation: Observe and evaluate the participant's creative use of objects in telling a novel story or creating a newscast or commercial.

ADOS-2

Fluent Speech Adolescent/Adult

Participant ID: _____

Gender: Female ☐ Male ☐

Date of Birth: _____

Date of Evaluation: _____

Chronological Age: _____

Examiner: _____

Other Information: _____

Observation/Coding

1. Construction Task*
2. Telling a Story From a Book
3. Description of a Picture*
4. Conversation and Reporting
5. Current Work or School*
6. Social Difficulties and Annoyance
7. Emotions
8. Demonstration Task
9. Cartoons*
10. Break
11. Daily Living*
12. Friends, Relationships, and Marriage
13. Loneliness
14. Plans and Hopes
15. Creating a Story

**optional*



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W-605E Test With Confidence

1 Construction Task *(optional)**Focus of Observation:*

- Does the participant indicate the need for more pieces?
 - If so, how does he or she attempt to do so?
(For example, does he or she reach over your arm?
Does he or she vocalize, gesture, or make eye contact?)
- Place the extra blocks out of reach, on the other side of your arm, so that the participant must reach over your arm to take the pieces (or ask for them).
- Show and tell the participant where the blocks are located, and initially encourage the participant to ask for more as needed.

Communication Sample:

2 Telling a Story From a Book*Focus of Observation:*

- Obtain a sample of the participant's spontaneous language and communication, as well as a sense of what captures his or her interest.
- Evaluate the participant's response to conventional humor, his or her spontaneous comments about how the characters in the story are feeling, and the degree to which he or she can convey continuity in a story.

3 Description of a Picture *(optional)*

Focus of Observation:

- Obtain a sample of the participant's spontaneous language and communication, as well as a sense of what captures his or her interest.

4 Conversation and Reporting

Focus of Observation:

- To what extent does the participant build on your statements, elaborate on his or her own statements to provide leads for you, and take a full role in back-and-forth conversation, particularly about a topic outside of the immediate context?
- How does the participant report *routine* and *nonroutine events*, and how does he or she describe *relationships* and *emotions*?
- Observe features of the participant's communication, including his or her use of gaze, facial expression, intonation, and gesture.

5

Current Work or School (optional)

If the participant is in school full time or does not have a job and has never been employed (including working as a volunteer), omit the employment questions and proceed to the questions about school.

Focus of Observation:

- Is the participant employed and/or in school?
- Has he or she had to leave jobs? If so, for what reasons?
- Does the participant have a realistic understanding of the possibilities for future employment?
- What has the participant's school experience been like?
 - Does he or she have a realistic understanding of the training or experience necessary for future employment?

Interview Questions About Work

- Do you have a job?

If so:

- What kind of job is it? How did you find it?
- Have you had other jobs before?
- Are you happy where you are or would you like to move on to something else eventually? What would it be?
- What about your co-workers? Do they seem to be happy or are they ready to move on?

If not:

- What do you do during the day?
- Did you have a job before? Why did you leave your old job? Was it something you had planned?
- Would you like to have a job someday?

If yes:

- What would it be?
- What will you need to do to find this type of job?

If no:

- What *would* you like to do?

OR

Interview Questions About School

- Are you in school? Where?
- What courses are you taking?
- What year (grade) are you in? How is it going?

If the participant is no longer in school and not employed:

- How far did you go in school? How did it go?
- What do you plan to do next? What experience/training would you need to do that?

6 Social Difficulties and Annoyance

Focus of Observation:

- Evaluate the participant's perception of social difficulties and his or her insight into the nature of these problems.
 - Has the participant made any attempt to change his or her own behavior in order to fit in with others more smoothly?
- Pay attention to the participant's understanding of the appropriateness and implications of his or her feelings and behaviors.

Interview Questions

- Have you ever had problems getting along with people at school or at work? How about at home? Do you ever get in trouble? Why? What for?
- Are there things that other people do that irritate or annoy you? What are they?
- What about things you do that annoy others?
 - (If no response, ask: What about your family members?)
- Have you ever been teased or bullied? Why, do you think?
- Have you ever tried to change these things? Have you ever done anything so that others wouldn't tease you? How has it worked?
- Are there other people you know who get teased or bullied?

Emotions

Focus of Observation:

- Identify the events or objects that elicit different emotions in the participant, particularly whether they are social in nature or not.
- Observe how the participant describes his or her emotions.
- Does the participant exhibit facial expressions or creative uses of language in the context of describing his or her emotions and others' emotions?
- Does the participant display insight into typical social relationships that may cause some of these emotions?

Interview Questions

- What do you like doing that makes you feel happy and cheerful?
- What kinds of things make you feel this way? How *do* you feel when you're happy? Can you describe it?
- What about things that you're afraid of?
- What makes you feel frightened or anxious? How does it feel?
What do you do?
- What about feeling angry?
- What kinds of things make you feel that way? How do you feel "inside" when you're angry?
- Most people have times when they feel sad. What kinds of things make you feel that way?
- How *do* you feel when you're sad? What is it like when you're sad?
Can you describe that?
- How about feeling relaxed or content? What kinds of things make you feel that way?



Demonstration Task

Focus of Observation:

- Does the participant represent familiar actions in gesture?
 - If so, how does he or she do this?
 - Does the participant use his or her body to represent an object (e.g., a finger for a toothbrush) or mime the use of a pretend object?
- Evaluate the participant's report of a routine event and the pragmatics of teaching a sequence of actions.



Cartoons (optional)

Focus of Observation:

- Observe the participant's use of gesture and its coordination with speech, as well as his or her response to humor.
- Obtain an additional language sample from the participant and a sense of his or her degree of flexibility in adapting a narrative to the audience of the listener.
- Note any comments the participant makes about emotions and relationships.



Break

Focus of Observation:

- How does the participant occupy himself or herself during free time?
 - How does he or she respond to your withdrawal from and return to the interaction?
- Does the participant initiate and participate in an unstructured conversation or interaction with you at the end of the break?
 - If so, how does he or she do this?

11 Daily Living *(optional)*

Focus of Observation:

- Obtain factual information about the participant's level of financial responsibility.
- Where is the participant living, and how was this arrangement made?
- To what extent is the participant realistic about plans for independence and the complexities of various living arrangements?

Interview Questions About Money

- What about money? Do you take care of your own money?
- Where does it come from? Who pays your bills?
- Have you ever saved your money to buy something or do something special? What was it?

Interview Questions About Residential Arrangements

- Where are you living now?

If living at home with parents:

- Have you ever lived away from your parents?
- What would be different about living on your own?
- Would you like it better?
- What would be difficult?

If living on his or her own:

- How did you find the place where you live now?
- Who do you live with?
- Can you tell me a little about it?

Interview Questions About Leisure Activities

- What do you like to do in your spare time at home?
- What about going out?

12 Friends, Relationships, and Marriage

Focus of Observation:

- Evaluate how the participant understands the concepts of friendship, marriage, and other social relationships, and the nature of these relationships.
- Evaluate the participant's understanding of why a person might want to be involved in a long-term relationship and of his or her own possible role in such a relationship.

Interview Questions

- Do you have some friends? Can you tell me about them?

(Note the ages of the friends. It can be helpful to ask for names if the participant is very general about who his or her friends are. Sometimes the names provided are family members or professionals or a list of an entire class, which is useful to know in terms of interpreting the answers to the questions below.)

- What do you like doing together? How did you get to know them?
How often do you get together?
- What does being a friend mean to you?
How do you know someone is your friend?
- How is a friend different from someone whom you just work with or go to school with?
- Do you have a girlfriend or boyfriend? What is her/his name?
How old is she/he?
- When did you see her/him last?
- What is she/he like? What do you like to do together?
- How do you know she/he is your girlfriend/boyfriend?
- Where do you want to live when you get older?
What kind of place (apartment, house, condo)?
- Whom do you think you would like to live with?
Your family, a roommate(s), by yourself?
- Do you ever think about having a long-term relationship or getting married (when you are older)?
- Why do you think some people get married or live with a girlfriend or boyfriend when they grow up?
- What would be nice about it? What might be difficult about being married or living with a girlfriend or boyfriend? Or living with a roommate?

13 Loneliness

Focus of Observation:

- Does the participant understand the concept of loneliness?
How does he or she feel it pertains to him or her or to other people?

Interview Questions

- Do you ever feel lonely?
- Do you think other people your age ever feel lonely?
- Are there things that you do to help yourself feel better?
What about things other people do to help themselves feel better when they're lonely?

14 Plans and Hopes

Focus of Observation:

- What does the participant anticipate in the future?

Interview Questions

- Do you have plans or dreams for the future of things that you would like to do, or have, or see happen to you? Tell me about them.

15 Creating a Story

Focus of Observation:

- Observe and evaluate the participant's creative use of objects in telling a novel story or creating a newscast or commercial.

CODING

- The overall ratings that you assign in this section should be made on the basis of the participant's behavior throughout the entire ADOS-2 administration.
- Ratings should include only behavior that is directly observed during the ADOS-2 administration and should not be based on behavior reported or observed in other contexts (e.g., parent report).
- If the participant's behavior changes in quality after a brief, initial adaptation period, ratings should be based on the period after the behavior stabilizes.
- Ratings should be assigned immediately after the ADOS-2 assessment.
- The ratings are organized according to five main groupings: "A. Language and Communication," "B. Reciprocal Social Interaction," "C. Imagination," "D. Stereotyped Behaviors and Restricted Interests," and "E. Other Abnormal Behaviors."

Language and Communication

Unless stated otherwise, code in relation to chronological age expectations, not in comparison to developmental level or estimated expressive language skills.

A1. Overall Level of Non-Echoed Spoken Language

The rating for this item should reflect the majority of the participant's utterances, not merely the best ones. For the purposes of the ADOS-2, a complex sentence is defined as an utterance with two or more clauses. Examples include "I didn't go to the zoo because it rained" or "I think wasps are really scary." In contrast, "I have two sisters and one brother" would not be considered a complex sentence.

0 = Uses sentences in a largely correct fashion (must use some complex speech).

1 = Some relatively complex speech (occasional utterances with two or more clauses), but with recurrent grammatical errors not associated with use of dialect.

2 = Non-echoed speech is mostly utterances of at least three words, but without complex language as described above.

3 = Non-echoed language is mostly simple phrases.

A2. Speech Abnormalities Associated With Autism (Intonation/Volume/Rhythm/Rate)

The focus of this item is on speech abnormalities that are characteristic of autism. Because of the variability within the autism spectrum, speech patterns in intonation, volume, rhythm, or rate (not articulation) that are unusual, but not obviously characteristic of autism, should receive a rating of 1. Code this item relative to the participant's expressive language level.

0 = Appropriately varying intonation, reasonable volume, and normal rate of speech, with regular rhythm coordinated with breathing.

1 = Little variation in pitch and tone; rather flat or exaggerated intonation, but not obviously peculiar, OR slightly unusual volume, AND/OR speech that tends to be somewhat unusually slow, fast, or jerky.

2 = Speech that is clearly abnormal for ANY of the following reasons: slow and halting; inappropriately rapid; jerky and irregular in rhythm (other than ordinary stutter/stammer), such that there is some interference with intelligibility; odd intonation or inappropriate pitch and stress; markedly flat and toneless ("mechanical"); consistently abnormal volume.

7 = Stutter or stammer or other fluency disorder (if odd intonation is also present, code 1 or 2 accordingly).

A3. Immediate Echolalia

This item pertains to the participant's immediate repetition of the last statement or series of statements made by the examiner or another person. When coding, do not include repetitions that are a lead-in to a response to the examiner or that are used as a memory device in specific tasks.

0 = Does not repeat others' speech.

1 = Occasional echoing.

2 = Echoing words and phrases regularly, but some spontaneous language, which can be stereotyped.

3 = Speech largely consists of immediate echolalia.

A4. Stereotyped/Idiosyncratic Use of Words or Phrases

Coding for this item includes delayed echolalia or other highly repetitive utterances with consistent intonation patterns. These words or phrases can be intended meaningfully and can be appropriate to conversation at some level. The focus of the item is on the stereotyped or idiosyncratic quality of the phrasing, unusual use of words or formation of utterances, and/or their arbitrary association with a particular meaning. Neologisms and referring to oneself by name should be coded here, as well as clear evidence of a pronoun error *across person* (e.g., *you* or *he* or *she* to mean *I*). Code relative to the participant's expressive language level.

0 = Rarely or never uses stereotyped or idiosyncratic words or phrases.

1 = Use of words or phrases tends to be more repetitive or formal than that of most individuals at the same level of expressive language, but not obviously odd, OR occasional stereotyped utterances or odd use of words or phrases, with substantial spontaneous, flexible language as well.

2 = Often uses stereotyped utterances or odd words or phrases, with some other language.

3 = Frequently uses odd or stereotyped speech, and rarely uses non-stereotyped spontaneous speech.

A5. Offers Information

The focus of this item is on the participant's spontaneous, appropriate offering of personal information, new to the examiner. It does not have to occur in context or be part of a sustained interaction. It can occur as the elaboration of responses to questions, but must include new information, not specified by the question. It can be related to the participant's interests, but should not be related solely to preoccupations. Comments about facts (e.g., "Did you know that whales are mammals?") are not coded here, but can be considered in assigning a rating under "Conversation" later in this section. Comments about relationships or possessions (e.g., "I have two brothers" or "Our family has a boat") can be coded here if they refer to an activity rather than a list of characteristics or objects. Lists of multiple characteristics (e.g., "I like to hike, sail, and fish") should be counted as one instance of offering information unless they are part of compulsive listing behavior, which does not receive credit here.

0 = Spontaneously offers information about his or her own thoughts, feelings, or experiences on several occasions.

1 = Occasionally offers information spontaneously about his or her own thoughts, feelings, or experiences.

2 = Rarely or never offers information spontaneously, except about circumscribed interests or preoccupations, OR offers information about facts or general knowledge, including preoccupations or circumscribed interests.

A6. Asks for Information

The focus of this item is on the participant's spontaneous expression of interest in the examiner's ideas, experiences, or reactions. This should not be part of a preoccupation. When assigning a rating, exclude asking for information that is not related to the examiner, or about the ADOS-2 materials, or about particular facts not specific to the examiner; instead, include these when assigning a rating under "Conversation." For this item, questions do not necessarily have to lead to a sustained conversation. Questions about relationships or possessions may be coded here if they refer to the examiner's experiences rather than filling in a list.

- 0** = Asks the examiner about his or her thoughts, feelings, or experiences on several occasions.
- 1** = Occasionally (at least one clear example) asks the examiner about his or her thoughts, feelings, or experiences.
- 2** = Responds appropriately to examiner's comments about his or her thoughts, feelings, or experiences, but does not spontaneously inquire about them.
- 3** = Rarely or never expresses interest in the examiner's thoughts, feelings, or experiences.

A7. Reporting of Events

The focus of this item is on the participant's ability to select an event spontaneously or in response to the examiner's general questioning, and to describe it in a comprehensible fashion without requiring specific probes. This should involve a sequential description of an event outside the immediate environment. Code the "best" example, given the rating constraints described below with regard to preoccupations and probes.

- 0** = Reports a specific nonroutine event (e.g., a holiday, a vacation, a shopping trip) that is not part of any preoccupations or intense interests and seems likely to be real. Gives a reasonable account without specific probes, but may need to be asked a general question to get started.
- 1** = Gives a reasonable account of a routine event (e.g., playing a favorite game, usual routine when he or she arrives home from work or school) that is not part of a preoccupation or intense interest and seems likely to be real. Gives the account without specific probes, but initially may need to be asked to describe the event. Include accounts from the "Demonstration Task" here.
- 2** = Provides an account of routine or nonroutine events, but dependent on specific probes, OR only describes an event that seems unlikely to have been real.
- 3** = Inconsistent or insufficient responses, even to specific probes.

A8. Conversation

This is a summary item that focuses on the to-and-fro use of words and phrases in social conversation. Code this item relative to the participant's expressive language level. Code evidence of (or lack of) *nonverbal* reciprocity under "Amount of Reciprocal Social Communication" in section B of this protocol. This rating should consider all opportunities for conversation, not merely the best.

0 = Conversation flows, building on the examiner's dialogue. This rating requires that much of the participant's speech provide both a response to the examiner's speech and some additional talking (not necessarily a question) that builds on what has just been said and allows a response from the examiner (i.e., sequences of at least four elements: examiner opens, participant comments, examiner responds, participant responds to response).

1 = Speech includes some spontaneous elaboration of the participant's own responses for the examiner's benefit OR provides leads for the examiner to follow, but either this is less in amount than would be expected for the participant's expressive language level or it is limited in flexibility.

2 = Little reciprocal conversation sustained by the participant; may follow his or her own train of thought rather than participate in an interchange; may have some spontaneous offering of information or comments, but little sense of reciprocity.

3 = Little spontaneous communicative speech (although there may be much echoed or noncommunicative speech). This rating can be used to describe participants who make some limited, but very few, responses to conversational initiations by the examiner.

A9. Descriptive, Conventional, Instrumental, or Informational Gestures

The focus of this item is on descriptive gestures that enact or represent an object or event (such as acting out rinsing a toothbrush or showing how a roller coaster curves through the air). Gestures that are conventional (e.g., clapping for "well done"), informational, or instrumental (e.g., pointing, shrugging, head nodding, or head shaking) receive partial credit. When coding, exclude emphatic gestures (e.g., "beats" accompanying speech, which are rated in the next item); include behaviors that occur during the "Demonstration Task" and throughout the ADOS-2 evaluation. The emphasis is on how the participant uses gestures before he or she is prompted or asked to do so, or gestures that the participant adds as he or she responds to a requested description (e.g., pretending to spit after demonstrating how to use a toothbrush, as requested). Pointing is included here as an instrumental gesture, as it is not coded separately in Module 4. Grabbing and reaching are not coded here.

0 = Spontaneous use of several descriptive gestures. These gestures may be typical or idiosyncratic, but must be communicative. May also use conventional or instrumental gestures.

1 = Some spontaneous use of descriptive gestures, but exaggerated or limited in range and/or contexts (e.g., occur only during "Demonstration Task"), OR frequent use of conventional or instrumental gestures, but rare or no use of descriptive gestures.

2 = Some spontaneous use of informational, conventional, or instrumental gestures, but rare or no use of descriptive gestures.

3 = No or very limited spontaneous use of conventional, instrumental, informational, or descriptive gestures.

8 = N/A (e.g., limited by physical disability).

A10. Emphatic or Emotional Gestures

The focus of this item is on emphatic (e.g., natural, rhythmic "beats" that often accompany speech) or emotional gestures (e.g., hand to mouth or hands up for "wow"). The rating for this item should be assigned on the basis of the timing of the gestures as they occur during speech. Other aspects of coordination of gesture, such as integration with gaze, should be coded under "Language Production and Linked Nonverbal Communication" or "Quality of Social Overtures" in section B.

0 = Variety of appropriate emphatic and/or emotional gestures that are well integrated with speech.

1 = Some emphatic or emotional gestures, but exaggerated, OR limited in frequency, appropriateness, integration, or style.

2 = Odd, excessive, or definitely awkwardly integrated emphatic or emotional gestures.

3 = No or very limited emphatic or emotional gestures.

8 = N/A (e.g., limited by physical disability).

B Reciprocal Social Interaction

Code in comparison to nonverbal mental age.

B1. Unusual Eye Contact

Coding for this item requires that clear, flexible, socially modulated, and appropriate gaze that is used for a variety of purposes be distinguished from gaze that is limited in flexibility, appropriateness, or contexts. If the participant is shy initially, and his or her gaze changes markedly and consistently as he or she becomes more comfortable, do not base the code on earlier impressions. However, if eye contact never improves, coding must be based on what is observed, even if the participant seems shy. Do not code eye contact that occurs between the participant and individuals other than the examiner who may be in the ADOS-2 assessment room.

0 = Appropriate gaze with subtle changes meshed with other communication.

2 = Uses poorly modulated eye contact to initiate, terminate, or regulate social interaction.

B2. Facial Expressions Directed to Examiner

The rating for this item should indicate whether the participant's facial expressions are directed to the examiner for the purpose of communicating affective (e.g., enjoyment, frustration) or cognitive (e.g., puzzlement, skepticism) states. Facial expressions that are directed to objects or other people in the room, or that are undirected, are not rated here. Appropriate or slightly exaggerated facial expressions should be coded even if there are also odd expressions.

0 = Directs a range of appropriate facial expressions to the examiner in order to communicate affective or cognitive states.

1 = Some direction of facial expressions to the examiner (e.g., directs only expressions indicating emotional extreme(s), or occasionally directs wider range of expressions). A participant who has a limited

B3. Language Production and Linked Nonverbal Communication

The purpose of this item is to code the degree to which, *when* the participant vocalizes, the vocalization is accompanied by subtle changes in gaze, facial expression, and gesture. This item should be coded on the basis of the vocalizations used, regardless of their frequency. Code the most typical occurrences, not merely the best ones. When assigning a rating, include vocalizations used to maintain interaction or to respond to the examiner, as well as initiations. A rating of 8 (uncodable) should be assigned by default if one or more of the following behaviors coded earlier in this protocol received a rating of 2: "Unusual Eye Contact," "Facial Expressions Directed to Examiner," or "Descriptive, Conventional, Instrumental, or Informational Gestures."

0 = Vocalization usually accompanied by subtle and socially appropriate changes in gesture, gaze, and facial expression.

1 = Vocalization accompanied by abnormal, limited, or less than usual frequency and/or range of gesture, gaze, and facial expression, OR use of one modality almost exclusively (e.g., frequent use of gaze, but limited use of gesture and facial expression).

2 = Little or no nonverbal communication linked with vocalization.

7 = Some avoidance of direct eye gaze, particularly at the beginning of the session, perhaps because of shyness, but shows some modulation and coordination of language and nonverbal behavior.

8 = N/A; no vocalization OR no or minimal use of gesture, facial expression, or socially directed gaze. This code should be assigned automatically if the absence of linking can be accounted for by the limited frequency of unusual eye contact, facial expressions,

B4. Shared Enjoyment in Interaction

Rate the participant's directed pleasure during any of the tasks or conversation. This item should not be used to indicate his or her general emotional state during the ADOS-2 evaluation. The rating applies to the participant's ability to *indicate* pleasure to the examiner, not just to interact or respond.

☐ 0 = Shows definite pleasure appropriate to context during interactive participation or conversation with the examiner in more than one task or conversational topic.

1 = Shows some pleasure appropriate to context during interactions with the examiner, OR shows definite pleasure during one interaction.

2 = Shows little or no expressed pleasure during interaction with the examiner, but shows pleasure in his or her own speech or actions or in noninteractive components of the ADOS-2 materials or activities.

3 = Little or no expressed pleasure during the ADOS-2 evaluation.

B5. Communication of Own Affect

The focus here is on the participant's ability to convey a range of his or her own emotions using words and facial expression, tone of voice, vocalization, and/or gesture. The coding for this item should reflect the range of emotion and the effectiveness of the communication (rather than the presence of specific emotions). Descriptions elicited during socioemotional questions may be rated here, as well as spontaneous comments and reports.

☐ 0 = Effective communication of a range of emotions that he or she is feeling or has felt.

1 = Some description of experiencing several emotions, but limited effectiveness of communication, AND/OR effective communication of at least one emotion.

2 = Some communication about at least one emotion.

3 = Minimal or no communication of his or her own affect.

B6. Comments on Others' Emotions/Empathy

The focus of this item is on the participant's communication of his or her recognition, understanding, and/or response to the feelings of other people or characters, real or conveyed in stories or other tasks. Exclude shared enjoyment with the examiner, which is rated in a preceding item.

☐ 0 = Spontaneously communicates clear understanding or labeling of and/or appropriate response to several different emotions in other people/characters. Labeling several emotions in others is sufficient but not necessary if there are other clear indications of understanding and/or appropriate response.

1 = Communicates some understanding, labeling, or response to an emotion in others (e.g., spontaneously and correctly identifies at least *one* emotion in another person/character).

2 = No or minimal identification/communication of understanding of emotion in others.

B7. Insight Into Typical Social Situations and Relationships

The focus of this item is on the participant's ability to provide spontaneous examples of insight into the nature of social relationships. These can include ongoing relationships, such as friendships or marriage, or interactive situations, such as getting along with other students or co-workers, that may be discussed in conversation or in response to the socioemotional questions. Two separate aspects of relationships are coded: (a) the nature of specific relationships (e.g., what is friendship), and (b) the participant's role in these relationships.

☐ 0 = Shows examples of insight into the nature of several typical social relationships (without evidence of lack of insight into those same relationships), including his or her own role in at least one. May show no more than one example of inaccurate understanding of other social relationships.

1 = Shows examples of insight into several typical social relationships, but not into his or her own role, OR into only one relationship including his or her own role.

2 = Shows some insight into one typical social relationship, though not necessarily about his or her own role in it.

3 = Shows no or limited insight into typical social relationships.

B8. Responsibility

This item focuses on the participant's references to and descriptions of being responsible for his or her own actions in typical daily living situations, in responding to normal social mores and expectations (from getting a haircut to finding a job) in an active, independent way.

☐

0 = Describes himself or herself as responsible for his or her own actions in several contexts, including dealing with minor problems in daily living.

1 = Provides at least one clear indication of being responsible for his or her own actions, but not consistent across contexts or consistently less straightforward than what is described above for a rating of 0. May show no more than one clear example of a lack of responsibility.

2 = Shows limited indication of a sense of responsibility for his or her own actions or shows more than one clear example of lack of responsibility that would be appropriate for his or her chronological age, taking into account developmental level.

B9. Quality of Social Overtures

This is a summary item that focuses on the *quality* of the participant's attempts to initiate social interaction with the examiner, not on the frequency of such attempts. Special attention should be given to the form of the overture and its appropriateness to the social context. The rating should reflect the majority of social overtures to the examiner, not merely the best ones.

☐

0 = Effectively uses nonverbal and verbal/vocal means to make clear social overtures to the examiner. The overtures must be appropriate to immediate contexts.

1 = Slightly unusual quality of some social overtures. Overtures may be restricted to personal demands or related to the participant's own interests, but with some attempt to involve the examiner in those interests.

2 = Inappropriate overtures; many overtures lack integration into context AND/OR social quality. This includes the participant's bringing up preoccupations with little attempt to involve the examiner in them.

3 = No social overtures of any kind.

B10. Amount of Social Overtures/Maintenance of Attention

The focus of this item is on the *number* of the participant's attempts to get, maintain, or direct the examiner's attention, AND/OR to direct the examiner's attention to objects, actions, or topics of interest to the participant. The rating for this item may include verbal or nonverbal behaviors if they are neither related to preoccupations nor aimed at getting objects, but seem to function primarily as a method of social contact. *Do not include requests when rating this item except for a code of 3.*

☐

0 = Frequent attempts to get or maintain the examiner's attention AND/OR to direct the examiner's attention to objects, actions, or topics of interest to the participant.

1 = Some attempts at getting, maintaining, or directing the examiner's attention as described above for a rating of 0, but reduced in frequency or the number of different activities in which they are used.

2 = Makes occasional attempts to get, maintain, or direct the examiner's attention, including overtures solely related to preoccupations.

3 = Shows relatively little concern as to whether the examiner is paying attention to him or her unless he or she needs help (e.g., initiates social contact only when requesting).

7 = Unusually frequent, intense, or excessive demands for attention.

B11. Quality of Social Response

This is a summary item that focuses on the participant's social responses throughout the ADOS-2 evaluation.

☐

0 = Shows a range of appropriate responses that are varied according to immediate social situations and presses.

1 = Shows responsiveness to most social contexts, but somewhat limited, socially awkward, inappropriate, inconsistent, or consistently negative.

2 = Odd, stereotyped responses, or responses that are restricted in range or inappropriate to the context.

3 = Minimal or no response to the examiner's attempts to engage the participant.

B12. Amount of Reciprocal Social Communication

The focus of this item is on the frequency with which reciprocal interchanges occur during the course of the ADOS-2 evaluation, using any mode of communication. *Frequency* here is defined both by absolute number of occurrences and distribution across a range of contexts. The rating for this summary item should describe aspects of nonverbal and verbal/vocal behavior that need not be coordinated but must result in at least brief reciprocal interchanges with the examiner (not others who may be present in the ADOS-2 assessment room).

- 0** = Extensive use of verbal or nonverbal behaviors (at whatever level attained) for social interchange (i.e., chat, comments, remarks, or nonverbal behaviors that appear to have reciprocal intent).
- 1** = Some reciprocal social communication (as described above for a rating of 0), but reduced in frequency or amount, or in the number of contexts in which such behaviors occur (regardless of the amount of nonsocial talk).
- 2** = Most communication is either object-oriented (i.e., to ask for things), OR response to questions, OR echolalic, OR concerned with particular preoccupations; little or no social chat or give-and-take.
- 3** = Little or no communication with the examiner.

B13. Overall Quality of Rapport

The code for this item is a summary rating that reflects the examiner's overall judgment of the rapport established with the participant during the ADOS-2 evaluation. The rating should particularly take into account the degree to which the examiner had to modify his or her own behavior to maintain the interaction successfully.

- 0** = Comfortable interaction between the participant and examiner that is appropriate to the context of the ADOS-2 assessment.
- 1** = Interaction sometimes comfortable, but not sustained (e.g., sometimes feels awkward or stilted, or the participant's behavior seems mechanical or slightly inappropriate).
- 2** = One-sided or unusual interaction resulting in a consistently mildly uncomfortable session or a session that would have been difficult if the examiner had not continuously modified the structure of the situation beyond the standard activities in the ADOS-2 evaluation.
- 3** = The participant shows minimal regard for the examiner, OR the session is markedly uncomfortable for a significant proportion of the time.

Imagination

Code this item in comparison to expressive language skills.

C1. Imagination/Creativity

This item should be assigned a rating that reflects the degree to which any of several forms of creativity/inventiveness are exhibited by the participant throughout the ADOS-2 evaluation, either in his or her use of objects or through verbal descriptions.

- 0** = Several different spontaneous, inventive, creative activities or comments in conversation.
- 1** = Some creative or make-believe actions, but limited in range or occurring only in response to one structured situation (e.g., creating a story).
- 2** = Little spontaneous creative or make-believe actions, OR only actions that are repetitive OR stereotyped in quality.
- 3** = No creative or inventive actions (not even stereotyped or repetitive).

D Stereotyped Behaviors and Restricted Interests

Code in relation to chronological age expectations.

D1. Unusual Sensory Interest in Play Material/Person

Rate the participant's interest in or unusual behaviors associated with sensory aspects of toys or surroundings (e.g., sniffing, repetitive feeling of texture, licking, mouthing, or biting, unusually strong interest in the repetition of certain sounds, unusual or prolonged visual examination).

If the participant has a preoccupation that is based on a sensory interest, this may be coded here as *one* unusual sensory interest. For example, if he or she shows an interest in radiators or plumbing, that is coded later in this section of the protocol under "D4. Excessive Interest in or References to Unusual or Highly Specific Topics or Objects or Repetitive Behaviors." If the participant is interested in the radiator in the room because he or she likes to look at it, as shown by peering at it while tilting his or her head, rocking from side to side, and jiggling his or her hands, this should be coded under "D2. Hand and Finger and Other Complex Mannerisms," but it may also be coded here because of the sensory component involved. If the participant likes to look out of the corner of his or her eye at the radiator, the corners of the room, the doors on the cabinets, and the slats of the window blinds, but does not become overly preoccupied with any of these objects and does not move in unusual ways as he or she does so, he or she should be coded here for unusual sensory interests but not under "Hand and Finger and Other Complex Mannerisms" or under item "D4. Excessive Interest..."

If the ADOS-2 assessment occurs in a room with a one-way mirror, looking into the mirror is not coded as an unusual sensory interest. Do not code here for touching the pin art. Sensory aversions are also not coded here.

- | | |
|--------------------------|--|
| ☐ | 0 = No unusual sensory interests or sensory-seeking behaviors. |
| ☐ | 1 = Several possible sensory interests not as clear as specified below for a rating of 2 AND/OR only one clear occurrence of an unusual sensory interest or a sensory-seeking behavior. One "possible" sensory interest should be coded 0. |
| ☐ | 2 = Definite interest in sensory elements of objects or of play materials, OR sensory examination of himself or herself or others; two or more clear occurrences must be observed. May be observed during the same activity. |
| ☐ | 3 = Definite unusual sensory-seeking behaviors occur frequently, during at least two different tasks or activities, and may interfere with the ADOS-2 assessment. |

Specify:

D2. Hand and Finger and Other Complex Mannerisms

Rate unusual and/or repetitive movements or posturing of the hands and fingers, arms, or body. Repetitive clapping may be coded here. Do not include body rocking unless it involves more than the torso. Finger tapping, nail biting, hair twisting, and thumb sucking are also not coded here. The participant does not have to watch the movements of his or her fingers or hands for the movements to be coded here.

- | | |
|--------------------------|---|
| ☐ | 0 = None. |
| ☐ | 1 = Unusual and/or repetitive hand and finger mannerisms or complex mannerisms not as clear as specified below for a rating of 2. |
| ☐ | 2 = Definite finger flicking or twisting, AND/OR hand or finger or complex mannerisms, stereotypes, or posturing. May be brief and/or rare if clear. |
| ☐ | 3 = Mannerisms, as described above, occur frequently, during at least two different tasks or activities, and/or may interfere with the ADOS-2 assessment. |

Specify:

D3. Self-Injurious Behavior

Rate behaviors that involve any kind of aggressive act to self, even if not clearly harmful.

- | | |
|--------------------------|--|
| ☐ | 0 = No attempts to harm self. |
| ☐ | 1 = Dubious or possible self-injury, and/or rare but clear self-injury (e.g., one clear example of biting at own hand or arm, pulling own hair, slapping own face, or banging own head). |
| ☐ | 2 = More than one clear example of self-injury, such as head banging, face slapping, hair pulling, or self-biting. |

D4. Excessive Interest in or References to Unusual or Highly Specific Topics or Objects or Repetitive Behaviors

Because circumscribed interests, preoccupations, or unusual behaviors are often difficult to judge during a brief observation, the focus of this item is on any references that (a) are unexpectedly high in frequency, (b) pertain to an unusual or odd topic, (c) are not well integrated into the conversation, or any behaviors that (d) pertain to use of an object in a manner highly specific to the participant, or (e) pertain to use of the participant's own body in a highly specific manner not clearly associated with behaviors coded under items D1 (Sensory Interests) or D2 (Hand/Finger Mannerisms); for example, putting his or her hands over and/or fingers in his or her ears should be considered here. Persistent aversive reactions that are unusual in form and/or intensity to sensory stimuli (e.g., the feel of the laminated picture, the sound of the examiner clearing his or her throat) can be coded here as 1, 2, or 3, as appropriate. Topics that are developmentally or age appropriate should not be coded here (e.g., a participant with a mental age of 8 years who repeatedly talks about a recent vacation in general terms would not be coded here; if he or she repeatedly talks about staying in hotel room 465, that behavior would be considered here).

The focus of this item is on the *topic* of references and/or unusual forms of behavior. Use of unusual terms (e.g., stereotyped phrases) and/or lack of conversational flexibility are coded elsewhere. Behaviors may be coded in two ways if they represent separate instances of each domain. For instance, if the participant repeatedly says "Do they need room service in room 465?", uses the same phrase in several other contexts ("No more room service!", "Room service now!"), and makes other statements about hotel room numbers, this would be coded both here and under "A4. Stereotyped/Idiosyncratic Use of Words or Phrases" earlier in this protocol. Repetitive behaviors involving objects that have a clear sequence (e.g., touching objects in a particular order) should be coded under the following item, "Compulsions or Rituals."

☐ 0 = No excessive interest in or references to unusual or highly specific or restricted topics or objects or repetitive behaviors.

☐ 1 = Occasional references to unusual or highly specific topics or patterns of interest, occurring to an unusual degree, or occasional repetitive behaviors.

☐ 2 = Definite, stereotyped, or unusual patterns of interest that may or may not intrude and/or interfere with social communication and/or definite repetitive behaviors.

☐ 3 = Definite preoccupation(s) and/or repetitive behaviors to a degree that interferes with the ADOS-2 assessment.

D5. Compulsions or Rituals

The emphasis for compulsions or rituals in this context is on the participant's determination to carry out an activity that involves a predictable sequence, endpoint, or manner that is not required as part of an ADOS-2 task (e.g., checking if a wallet is in a purse; insistence on completing the book used for the storytelling task; careful placement of materials as they were initially presented; reciting a list of classmates as friends). Provision of lists should be rated here.

☐ 0 = No obvious activities or verbal routines that must be completed in full or according to a sequence that is not part of the task.

☐ 1 = Unusually routinized in speech or activities (includes insistence on finishing the book or providing a list that is not relevant to the conversation), but no behavior that appears clearly compulsive in quality.

☐ 2 = One or more activities or verbal routines that the participant has to perform or say in a special way. The participant appears under pressure or becomes anxious if an activity is disrupted (i.e., compulsive quality is present). Include the recitation of lists that must be completed or that the examiner is asked to record (e.g., friends, favorite foods) or insistence that the examiner respond in a specific way.

Specify:

Other Abnormal Behaviors

Unless stated otherwise, code these items without reference to developmental level or estimated expressive language skills.

E1. Overactivity/Agitation

This item describes excessive movement or physical agitation.

☐ 0 = Sits still appropriately throughout the ADOS-2 assessment.

1 = Sits, but often fidgets or moves about in the chair. Difficulties in the ADOS-2 assessment are not principally due to overactivity or agitation.

2 = Difficulty sitting; moves either in or out of the chair or handles or manipulates objects in a way that is mildly disruptive.

3 = Overactive behaviors are difficult to interrupt. The level of activity disrupts the ADOS-2 assessment.

7 = Underactive.

E2. Tantrums, Aggression, Negative or Disruptive Behavior

This item includes any form of anger or disruption beyond communication of mild frustration or whining.

☐ 0 = Not disruptive, destructive, negative, or aggressive during the ADOS-2 assessment.

1 = Displays an example of mild disruption, anger, or aggression or negative behavior to the examiner (includes verbal threats, swearing, or a deliberately loud voice).

2 = More than one intentionally disruptive or negative incident. Loud talking or repeated swearing is coded here.

3 = Shows marked or repeated temper tantrums or significant aggression (e.g., throwing things, hitting or biting others). Screaming or yelling is included here.

E3. Anxiety

Anxiety includes initial wariness or self-consciousness, as well as more obvious signs of worry, upset, or concern.

☐ 0 = No obvious anxiety (e.g., trembling or jumpiness).

1 = Mild signs of anxiety or self-consciousness, especially at the beginning of the ADOS-2 session or in response to specific activities.

2 = Marked anxiety throughout the ADOS-2 assessment (may be intermittent or continuous).

Participant ID: _____ Examiner: _____
 Gender: Female ☐ Male ☐ Date of Birth: _____ Date of Evaluation: _____ Chronological Age: _____

CONVERTING ITEM CODES TO ALGORITHM SCORES

- Convert assigned ratings of 3 to algorithm scores of 2.
- Convert assigned ratings other than 0, 1, 2, or 3 (i.e., 7, 8, and 9) to algorithm scores of 0.
- Transfer assigned ratings of 0, 1, and 2 directly to the algorithm form (do not convert).

Communication

Stereotyped/Idiosyncratic Use of Words or Phrases (A-4)
 Conversation (A-8)
 Descriptive, Conventional, Instrumental, or Informational Gestures (A-9)
 Emphatic or Emotional Gestures (A-10)

COMMUNICATION TOTAL

Reciprocal Social Interaction

Unusual Eye Contact (B-1)
 Facial Expressions Directed to Examiner (B-2)
 Comments on Others' Emotions/Empathy (B-6)
 Responsibility (B-8)
 Quality of Social Overtures (B-9)
 Quality of Social Response (B-11)
 Amount of Reciprocal Social Communication (B-12)

SOCIAL INTERACTION TOTAL

COMMUNICATION + SOCIAL INTERACTION TOTAL

See the back of this form for guidance on how to convert the Communication Total, Social Interaction Total, and Communication + Social Interaction Total to the ADOS-2 Classification.

Imagination/Creativity

(C-1)

Stereotyped Behaviors and Restricted Interests

Unusual Sensory Interest in Play Material/Person (D-1)
 Hand and Finger and Other Complex Mannerisms (D-2)
 Excessive Interest in Unusual or Highly Specific Topics/Objects or Repetitive Behaviors (D-4)
 Compulsions or Rituals (D-5)

STEREOTYPED BEHAVIORS AND RESTRICTED INTERESTS TOTAL

CLASSIFICATION/DIAGNOSIS

ADOS-2 Classification: _____

Overall Diagnosis: _____

Converting the Communication Total, Social Interaction Total, and Communication + Social Interaction Total to the ADOS-2 Classification

Compare the Communication Total, Social Interaction Total, and Communication + Social Interaction Total to the cutoff scores below for Module 4:			
	COMMUNICATION	SOCIAL INTERACTION	COMMUNICATION + SOCIAL INTERACTION
autism	3	6	10
autism spectrum	2	4	7

Assign the ADOS-2 Classification:

autism	All three totals (Communication Total, Social Interaction Total, combined Communication + Social Interaction Total) are equal to or greater than the three separate corresponding <i>autism</i> cutoffs: ■ All of the following are true: Communication Total is 3 or higher AND Social Interaction Total is 6 or higher AND Communication + Social Interaction Total is 10 or higher
autism spectrum	All three totals (Communication Total, Social Interaction Total, combined Communication + Social Interaction Total) are equal to or greater than the three separate corresponding <i>autism spectrum</i> cutoffs, but at least one is less than its corresponding <i>autism</i> cutoff: ■ All of the following are true: Communication Total is 2 or higher AND Social Interaction Total is 4 or higher AND Communication + Social Interaction Total is 7 or higher AND ■ At least one of the following is also true: Communication Total is 2; Social Interaction Total is 4 or 5; Communication + Social Interaction Total is 7 to 9
non-spectrum	Any one of the three totals (Communication Total, Social Interaction Total, combined Communication + Social Interaction Total) is less than the <i>autism spectrum</i> cutoff: ■ At least one of the following is true: Communication Total is 1 or lower; Social Interaction Total is 3 or lower; Communication + Social Interaction Total is 6 or lower

Record the ADOS-2 Classification on the front of this form in the space marked ADOS-2 Classification.